

Acute Mountain Sickness (AMS)

Background - Usually only occurs with altitude 7000ft to 8000ft (2000m) - May occur at lower altitudes in susceptible population (COPD, CHF, obesity, history of AMS) - Development of symptoms based on: - rate of ascent - sleeping altitude - strength of hypoxic ventilatory response - alcohol intake - obesity - NOT based on physical fitness, age, sex, smoking, previous high-altitude experience - Tend to have recurrence of symptoms whenever they return to the symptomatic altitude - Low partial pressure of O₂ leads to decreased diffusion of O₂ across alveolar spaces --> hypoxemia results in tachypnea and cerebral vasodilation (↓CO₂) --> increased ICP	Physical Lake Louise Consensus Definition - Individual is at or above 2000m (7000ft) above sea level AND - A headache is present AND - The individual has any one of the following: - GI Symptoms (nausea, vomiting, anorexia) - Sleep symptoms (insomnia, difficulty sleeping) - Dizziness and/or lightheadedness Timeline Onset - Symptoms usually develop 1-6hr after arrival at elevation - May be delayed for 1-2d - Especially common after the 1st or 2nd night's sleep Duration - Average duration of symptoms at 10,000ft =15hr - At higher elevations symptoms may last weeks / more likely to progress to headache
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EMT/AEMT

- Follow General Medical Care Guideline
- Perform a full neurologic exam
- If neuro and pulmonary exam is negative, provide education on prevention of HAPE/HACE
 - Stay at current or lower altitude until symptoms have resolved and patient no longer requires symptomatic treatment (12-36 hours)
 - Stay well hydrated/Avoid alcohol use
 - Sleeping altitude is most important - acclimatization hikes (after improvement of symptoms) are warranted as long as sleeping elevation remains lower than elevation where symptoms began
- If symptoms improve in 12-36 hours, may continue to ascend cautiously
- If symptoms return, repeat above measures and contact medical director for advice
- Descend immediately if neurologic deficit or signs of HAPE**
- Descend if symptoms worsen or fail to improve within 1-2 days of effective therapy for symptoms**

EMT/AEMT - Expanded Scope

- Pain management per Pain Management Guidelines
- Zofran 4mg IM/ODT every 5 min x 2 doses (max 8mg)**

Paramedic

- Follow General Medical Care Guideline
- Toradol 15mg IM/IV or Ibuprofen 400mg every 6 hours for headache**
- Zofran 4mg ODT/IM/IO/IV every 5 min x 2 doses (max 8mg)**
- If the patient has **Acetazolamide** - may advise begin taking Acetazolamide 250mg twice daily
 - Contraindications: Anaphylaxis to sulfonamide medications, Hx Stevens-Johnson Syndrome
 - Advise of common side effects - bilateral paresthesia in hands/feet, loss/change of taste (not smell), polyuria (worsening dehydration)
 - Side effects are not a reason to stop the medication - they will be reversible when stopping the medication
 - It is important to stay hydrated and eat adequate calories despite loss/change of taste

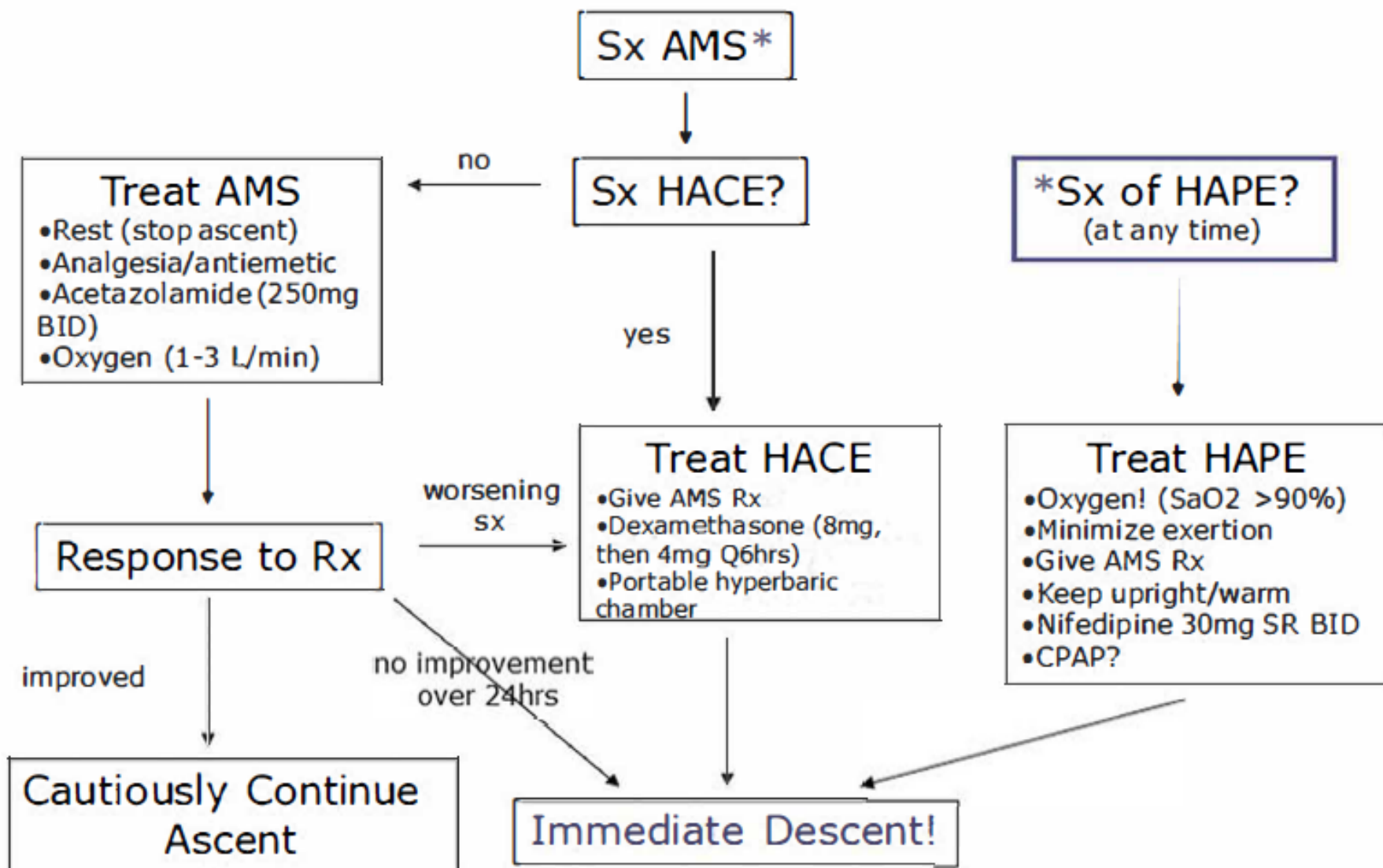


Table 2 Risk categories for acute mountain sickness

<i>Risk category</i>	<i>Description</i>
Low	• Individuals with no history of altitude illness and ascending to :52800 m
	• Individuals taking 2:2 d to arrive at 2500-3000 m with subsequent increases in sleeping elevation <500 m-d⁻¹ and an extra day for acclimatization every 1000 m
Moderate	• Individuals with history of AMS and ascending to 2500-2800 m in 1 d
	• No history of AMS and ascending to >2800 m in 1 d
	• All individuals ascending >500 m-d⁻¹ (increase in sleeping elevation) at altitudes above 3000 m but with an extra day for acclimatization every 1000 m
High	• Individuals with a history of AMS and ascending to >2800 m in 1 d
	• All individuals with a history of HACE or HAPE
	• All individuals ascending to >3500 m in 1 d
	• All individuals ascending >500 m-d⁻¹ (increase in sleeping elevation) above >3000 m without extra days for acclimatization
	• Very rapid ascents (eg, <7 d ascents of Mt. Kilimanjaro)

AMS, acute mountain sickness; HACE, high altitude cerebral edema; HAPE, high altitude pulmonary edema.

Altitudes listed in the table refer to the altitude at which the person sleeps.

Ascent is assumed to start from elevations <1200 m.

The risk categories described pertain to unacclimatized individuals.